MIL-S-50004A(MU)
24 June 1968
SUPERSEDING
MIL-S-50004(Ord)
21 November 1958

MILITARY SPECIFICATION

SODIUM SULFATE, ANHYDROUS, TECHNICAL (FOR AMMUNITION USE)

1. SCOPE

1.1 This specification covers sodium sulfate used for manufacture of propellants.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

UU-S-48 - Sacks, Paper, Shipping
PPP-B-35 - Bags, Textile, Shipping, Burlap

STANDARD

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 6810

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2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station. 516 West Jackson Boulevard, Chicago, Illinois 60606.)

INTERSTATE COMMERCE COMMISSION

49 CFR 71-78 - Interstate Commerce Commission Rules and Regulations for the Transportation of Explosives and Other Dangerous Articles

(The Interstate Commerce Commission regulations are now a part of the Code of Federal Regulations (1949 Edition-Revised 1950) available from the Superintendent of Documents, Government Printing Office, Washington, D. C. 20402. Orders for the above publication should cite "49 CFR 71-78 (Rev 1950)").

3. REQUIREMENTS

3.1 Physical appearance. The anhydrous sodium sulfate shall be in the form of a white powder.

3.2 Moisture content. The moisture content of sodium sulfate shall not exceed 0.5 percent when tested as indicated in 4.5.2.

3.3 Assay. The anhydrous sodium sulfate shall be not less than 98.0 percent as sodium sulfate when tested as indicated in 4.5.3.

3.4 Neutrality. The acid shall be not more than 0.005 percent calculated as sulfuric acid, or the alkalinity shall be not more than 0.5 percent calculated as sodium carbonate when tested as indicated in 4.5.4.

3.5 Halides. The halide content as sodium chloride (NaCl) shall not exceed 0.7 percent when tested as indicated in 4.5.5.

3.6 Insoluble matter. The insoluble matter shall not exceed 0.1 percent when tested as indicated in 4.5.6.

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3.7 Workmanship. Workmanship shall be in accordance with high-grade commercial practice covering the manufacture of anhydrous sodium sulfate in order to obtain a product suitable for Army use. The anhydrous sodium sulfate shall be free from lumps, caking, and visible impurities.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Contractor quality assurance system. The contractor shall provide and maintain an adequate quality assurance system, acceptable to the Government covering the materials under the contract. A current written description of the system shall be submitted to the contracting officer prior to initiation of production. The written description will be considered acceptable when, as a minimum, it provides the quality assurance required by this specification. The contractor shall notify the Government of, and obtain approval for, any change to the written procedure that might affect the degree of assurance required by this specification or other applicable documents referenced herein.

4.2 Lot. Unless otherwise specified, a lot shall consist of all the anhydrous sodium sulfate supplied at the same time, which has been produced by one manufacturer under the same manufacturing conditions, provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 4.3). If a lot consists of less than 5 containers, every container shall be sampled.

4.3 Batch. A batch shall be that quantity of the chemical manufactured at the same time. When practical, all anhydrous sodium sulfate supplied on one order shall be from the same manufacturer's batch. If in any one delivery the chemical is from more than one batch, the batch number shall be on the label.

4.4 Sampling. For chemical tests, 1/4 pound representative specimen shall be removed from each of five containers taken at random from the lot. The specimen shall be placed in a clean, dry container, thoroughly blended, and labeled to identify the container and lot represented.

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4.5 Test methods.

4.5.1 General. Analytical reagent grade chemicals and distilled water shall be used throughout the tests. Blank determinations shall be run in parallel with the tests, using the same quantities of reagents used in the test, and corrections shall be applied when necessary.

4.5.2 Moisture. Transfer a 2.00 gm. sample to a tared weighing bottle. Dry the bottle and contents at 150°C for 1 hour, cool in a desiccator, and weigh. Calculate the percent moisture as follows:

$$\text{Percent moisture} = \frac{100A}{W}$$

where: A = loss in weight, gm.
W = weight of sample, gm.

4.5.3 Assay. Take about 0.5 gm. of the sample used for the moisture determination and weigh to the nearest 0.1 mg. Dissolve in 400 ml. of water containing 4 ml. of hydrochloric acid. Heat to boiling and add dropwise 10 ml. of barium chloride solution (10 percent). Heat for 2 hours on the steam bath. Filter through a tared sintered porcelain crucible and wash with water. Ignite at 700°C gently for 1 hour, cool in a desiccator, and weigh. Calculate the percent sodium sulfate as follows:

$$\text{Percent sodium sulfate} = \frac{60.86A}{W}$$

where: A = weight of precipitate, gm.
W = weight of sample, gm.

4.5.4 Neutrality. Dissolve a 10.00 gm. sample in 200 ml. of carbon dioxide free water and add methyl red indicator. If the solution is acid, titrate to a change in color with 0.01 N sodium hydroxide solution and calculate the acidity as follows:

$$\text{Percent sulfuric acid} = \frac{4.9\ NV}{W}$$

where: N = normality of sodium hydroxide solution
V = volume of sodium hydroxide solution
W = weight of sample, gm.

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If the solution is basic, titrate with 0.01 N hydrochloric acid and calculate the alkalinity as follows:

$$\text{Percent sodium carbonate} = \frac{5.3\ NV}{W}$$

where: N = normality of hydrochloric acid
V = volume of hydrochloric acid
W = weight of sample, gm.

4.5.5 Halides. Dissolve a 10.00 gm. sample in 200 ml. of water and filter. Add 1 ml. of nitric acid and heat to boiling. Add 5 ml. of silver nitrate solution (10 percent) and boil for a minute. Allow to stand overnight in a dark place. Filter through a tared sintered glass crucible. Wash with dilute nitric acid (1 percent) and dry at 150°C for 1 hour. Cool in a desiccator and weigh. Calculate the percent sodium chloride as follows:

$$\text{Percent sodium chloride} = \frac{40.79A}{W}$$

where: A = weight of precipitate, gm.
W = weight of sample, gm.

4.5.6 Insoluble matter. Dissolve a 10.00 gm. sample in 200 ml. of water, filter through a tared sintered glass crucible, and wash with water. Dry at 105°C for 1 hour, cool in a desiccator and weigh. Calculate the percent insoluble matter as follows:

$$\text{Percent insoluble matter} = \frac{100A}{W}$$

where: A = weight of insoluble matter, gm.
W = weight of sample, gm.

4.6 Resubmission and retest. If the composite sample, or any primary sample subjected to test, fails to pass any of the tests, the lot shall be rejected. The contractor shall have the option of having a partial or complete analysis made on samples taken from any or all of the containers in the lot at no expense to the Government. The contractor may then remove the defective portion of the lot, and resubmit the lot for acceptance provided complete replacement of the defective portion can be made to the satisfaction of the inspector. The resubmitted lot shall be accepted provided the new samples selected in accordance with 4.4 pass all the required tests.

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5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level A. Unless otherwise specified, one hundred pounds of sodium sulfate shall be packed in a new full removable head metal drum meeting the requirements of ICC specification 37A or in a textile and paper laminated shipping bag in accordance with PPP-B-35, type III, class 2, construction number P12 or P15.

5.1.2 Level B. Unless otherwise specified one hundred pounds of sodium sulfate shall be packed in a 5-ply multi-wall bag, type 4W, type optional, meeting the requirements of UU-S-48.

5.1.3 Level C. The sodium sulfate shall be packed in accordance with the manufacturer's standard commercial practice in a manner which will provide protection against damage during shipment from the source of supply to the first receiving activity for immediate use. Shipping containers used shall conform with Uniform Freight Classification Rules or other common carrier regulations applicable to the mode of transportation.

5.2 Marking. In addition to any special marking specified by contract or order, shipments shall be marked in accordance with MIL-STD-129.

.6. NOTES

6.1 Intended use. Sodium sulfate is intended for use in hardening process of smokeless propellants.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification
(b) Applicable level of packing.
(c) Size and type of container required (see 5.1)

Custodian:
Army - MU

Preparing activity:
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Review interest:
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User interest:
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